# Aditya Venkata Subramanian (Aditya V)

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Aditya V Aditya-Venkat aditya-venkat.github.io

## Education

**BS** Indian Institute of Technology Kanpur, Statistics and Data Science — 2022 – 2026
**CPI:** 9.4/10

- **Awards:** Academic Excellence Award (2022, 2023 & 2024)
- **Honors Track Student**
- **Minor:** Machine Learning and Applications

**Grade XII** National Centre for Excellence, Bengaluru — **Score:** 97.2% | 2022

**Grade X** National Centre for Excellence, Bengaluru — **Score:** 97.2% | 2020

## Preprints

*Links to below papers will be updated on my personal website as they become available*

**Benchmarking of Spatial Transcriptomics Methodologies**
Ajita Shree, Aditya V, Tanush Kumar, Hamim Zafar (will be available soon on biorxiv)
Comprehensive benchmarking of 20+ spatial transcriptomics methods across simulated datasets, introducing novel metrics and simulation techniques. Poster accepted at RECOMB 2025, Seoul

**A Bayesian Bivariate Load-sharing Model**
Aditya V, Debasis Kundu (Manuscript under preparation)
Proposed a Bayesian extension of classical load-sharing models with multiple baseline distributions and MCMC-based credible interval estimation for reliability parameters

## Research Experience

**Benchmarking of Spatial Transcriptomics Methodologies** | Prof. Hamim Zafar | IIT Kanpur (*Sep '23 - Nov '25*)
*Second-authored a poster accepted to RECOMB 2025, Seoul*

- Built pipelines of **10+** different Spatial Deconvolution methodologies and **10+** Spatial Domain detection methods
- Evaluated the methods on **35+** simulated datasets from **4+** different categories on over **80 Gigabytes** of data
- Designed a **novel spatial bivariate statistic** and implemented **simulation methods** for **imaging-based data**
- Developed essential, consistent insights of method performance using **10+** novel, standard and spatial metrics

**Bayesian Bivariate Load-sharing Model** | Prof. Debasis Kundu | IIT Kanpur | (*Jan '25 - present*)

- Implemented a **bivariate load-sharing model**, overcoming flaws in traditional models and tested estimation approaches on simulated datasets
- Created a novel **Bayesian extensions** for exponential, weibull and piecewise constant hazard baselines using suitable priors, and demonstrated usability
- Implemented **MCMC sampling techniques** to estimate credible intervals of parameters of interest

**ML and Cancer Research Internship**| Dr. Karthik | TenSixty Bio | Massachusetts (remote) (*Jun '24 - Aug '24*)

- Worked with the co-founders to build the **spatial binding** component of their cancer therapeutics pipeline
- Built an **unsupervised Computer Vision algorithm** to quantify IHC staining at cellular resolutions
- Used **clustering-based approaches**, leveraging **cell-segmentation models** to measure antibody expression
- Set up a process hosted on **Amazon Web Services** to extract and analyze antibody expression across samples

## Selected Projects

**Sampling-based Bayesian Inference Unlearning** | MTH707 | Prof. Dootika Vats | IIT Kanpur | (*Jan '25 - Apr '25*)

- Investigated **MCMC-based machine unlearning** algorithms for **Bayesian inference models** without full retraining
- Explored **KL-divergence** based inference unlearning and potential for a novel extension to **JSD-based unlearning**

- Designed methods using **low-rank Hessian approximations** to improve unlearning accuracy by integrating **Stochastic Gradient Reimann Langevin Dynamics** with KL-divergence based unlearning
- Conducted **simulation studies** validating posterior recovery, and error bounds under data removal scenarios

**Multimodal Cell Type Imputation** | Prof. Hamim Zafar | IIT Kanpur | (*Sep '25 – present*)

- Built **spatial joint embeddings** using GAN-based models conditioned with **niche-based inference parameters**
- Applied **non-parametric classifiers** (Spearman correlation, multi-class p-tests) for cell-type inference
- Developing a **graph-based dual-autoencoder with adversarial training** for improved spatial representations.

## Technical Skills

**Languages:** Python, C, C++, R, SQL **Other Tools:** Git, Markdown, AWS, NextFlow, RShiny, LaTeX
**Libraries:** Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn, HuggingFace, PyTorch, OpenCV, XML.eTree

## Relevant Coursework

*Legend:* ⋆⋆ Outstanding Performance ⋆ Excellent Performance † Ongoing Course

| Foundational courses | Methodology & Modeling | Computational Data Science |
|---|---|---|
| Introduction to Probability Theory | Linear Regression and ANOVA ⋆ | Introduction to Machine Learning ⋆⋆ |
| Matrix Algebra & Linear Estimation | Non-Linear Regression ⋆⋆ | Data Science Labs ⋆ |
| Theory of Statistics | Time Series Analysis ⋆ | Statistical Computing |
| Stochastic Processes ⋆ | Non-Parametric Inference † | Data Structures and Algorithms |
| Single & Several Variable Calculus ⋆ | Introduction to Bayesian Analysis | Fundamentals of Computing ⋆ |
| Ordinary Differential Equations ⋆ | Markov Chain Monte Carlo ⋆ | Computational Genomics † |

## Scholastic Achievements

- Secured **All India Rank 817** in JEE Advanced & **All India Rank 721** in JEE Mains among 1 Million candidates *2022*
- Secured **All India Rank 3** in IISER Aptitude test amongst 20 thousand candidates from across the country *2022*
- Secured **All India Rank 641** in KVPY SA amongst 50 thousand candidates and became eligible for scholarship *2021*

## Mentorship

**Sound Refine** | Project Mentor | BCS, IIT Kanpur | (*May '24 - July '24*)

- Engineered a pipeline for **noise reduction**, **transcription**, and **summarization** on data built via audio mixing
- Led mentees through the basics of Python, Machine Learning, Audio processing, **CNNs**, **U-Nets**, and **Transformers**

**Academic Mentor, PHY115** | Counselling service, IIT Kanpur (*Apr '23 - Apr '24*)

- Guided **600+** freshers through a course on oscillation and waves and was a personal mentor to **16** students
- Delivered a lecture to a classroom of **300** students on the wave equation and Fourier transforms for their end-term

## Work Experience

**Associate Consulting Intern** | Bain & Co. | Mumbai (Onsite) (*May '25- Jul '25*)

- Designed **automated models** to recommend organizational restructuring actions across **40+ client accounts**
- Developed **quantitative techniques** to estimate value creation potential ("size of prize") under limited data
- Optimized account-level overheads by **clustering 40+ accounts**, identifying cost drivers, and benchmarking

## Positions of Responsibility

**Coordinator** |Debating Society, IITK (*Apr '24 - Apr '25*)

- Initiated and convened the first and second editions of IITKPD with **80+** participants, and a budget of **INR 2,50,000**
- Organised **UPDL'24** at IIT Kanpur, a national **3-tier** tournament with **3000+** participants from over **8** states
- Improved debate training quality by introducing external debate coaches and brought in funds over **INR 1,00,000**